INCH-POUND
MIL-DTL-10296K
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SUPERSEDING
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12 September 1986

DETAIL SPECIFICATION

CLOTH, SATEEN, COTTON

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers a cotton sateen cloth.

1.2 Classification. This specification will cover one type of cloth in the following classes:

1.2.1 Classes.

Class 1 - Dyed
Class 2 - White

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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AMSC N/A FSC 8305

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52094 – Thread, Cotton

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

FEDERAL TRADE COMMISSION

16 CFR Part 303 Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies of this document are available online at https://www.ftc.gov.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	- Evaluation Procedure for Gray Scale for Color Change
AATCC EP2	- Evaluation Procedure for Gray Scale for Staining
AATCC EP8	- Evaluation Procedure for AATCC 9-Step Chromatic transference Scale
AATCC EP9	- Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8	- Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM15	- Test Method for Colorfastness to Perspiration
AATCC TM16.3	- Test Method for Colorfastness to Light: Xenon Arc
AATCC TM61	- Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81	- Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM96	- Test Method for Dimensional Changes in Commercial Laundering of Woven and Knitted Fabrics Except Wool

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(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D1683/D1683M	- Standard Test Method for Failure in Sewn Seams of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

Fabric Defect Replica Scales

(For Information on obtaining copies of the Fabric Defect Replica Scales please contact U.S. Army Combat Capabilities Development Command (DEVCOM), Soldier Center, ATTN: FCDD-SCD-PE, 10 General Greene Avenue, Natick, MA 01760-5019.)

INFORMA HEALTHCARE

Repeat Insult Patch Test - Modified Draize Procedure –
Principles and Methods of Toxicology, A Wallace Hayes (editor).

(Copies of this document are available online at https://www.crcpress.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.2 Standard sample. Unless otherwise indicated, the finished cotton, sateen cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.2).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Figure. Figure 1 is furnished for informational purposes only. To the extent of any inconsistencies between the written document and the figure, the written document shall govern.

3.5 Materials.

3.5.1 Yarn. The yarn shall be made from carded cotton that has been drawn and spun into single yarns.

3.5.2 Weave. The weave shall be 5 harness sateen. The filling effect side of the cloth shall be of the face side, and the weave shall be as shown on Figure 1 when viewed from the face side.

3.6 Color.

3.6.1 Class 1. The color for class 1 cloth shall be as specified in the contract or purchase order (see 6.2).

3.6.2 Class 2. The color for Class 2 cloth shall be bleached white. The cloth shall be bleached using a peroxygen type agent and a fluorescent optical brightener with a resulting peak emission in the blue violet.

3.6.3 Visual shade matching. The color and appearance of class 1 and 2 cloths shall match the standard sample (see 6.3) when tested as specified in 4.6.2.

3.6.4 Labile sulfur. Dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid shall not be used. The dyed class 1 cloth shall be “Free” of labile sulfur when tested as specified in 4.6.1.

3.6.5 Colorfastness. Unless otherwise specified in the contract or procurement documents, the finished cotton sateen cloth referenced in 3.6 shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

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TABLE I. Colorfastness requirements.

Classes	Laundering (Color Change and Staining) (3 cycles) (min.)	Light (after 40 AFU or 170 kJ/(m^2nm) @420nm) (min.) 1/, 2/	Perspiration (Acid & Alkaline) (Color Change and Staining),(min.)	Crocking (Dry/Wet) (min.)
Class 1	4	4	4	4
Class 2, 3/	4	4	4	--

1/ AFU: AATCC Fading Units
2/ For Class 2 (only), the exposure shall be 20 AFU or 85 kJ/(m^2nm) @420nm).
3/ Evaluated for color change only.

3.7 Physical requirements. The finished cotton, sateen cloth shall conform to the requirements, listed below, in Table II, when tested as specified in 4.5.

Table II. Physical requirements (all classes)

Characteristic	Requirements	
	Class 1	Class 2
Weight, oz./sq.yd. Minimum	9.0	9.0
Yarns per inch, (minimum) Warp Filling	 85 48	 85 48
Breaking strength (pounds), (minimum) Warp Filling	 115 100	 100 90

3.8 Finish. The Class 1 cotton, sateen, cloth shall be singed, desized, and mercerized when tested as specified in 4.5.

3.9 pH. The pH of the water extract of the finished cotton, sateen cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.10 Dimensional stability. The shrinkage or elongation both in the warp and filling of the finished cotton, sateen, cloth shall not be greater than 2.0 percent in either direction when tested as specified in 4.5.

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3.11 Seam efficiency. The finished cotton, sateen cloth shall have a seam efficiency of not less than 80 percent when tested as specified in 4.5.

3.12 Toxicity. The finished cotton, sateen, cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.6.3. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used.

3.13 Face/back identification. The warp side shall be identified as the face side by stamping that side with the word "FACE" or "Back" at each end of the roll.

3.14 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width inclusive of the selvage when fly-shuttle looms or shuttleless looms with tuck-in selvage are used. For all other shuttleless looms, the width measurement shall be made between the last warp thread on each side excluding the protruding fringe (s).

3.15 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put-up full width on a roll.

3.16 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.17 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the quality acceptance levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article, submitted in accordance with 3.1, shall be inspected, examined for appearance, color, and finished defects listed in 4.4 and tested for the characteristics as specified in 4.5.

4.3 Conformance inspection. Conformance inspection shall include the visual examination of 4.4 and the tests of 4.5 through 4.6 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with acceptance quality limits (AQLs) as specified in the contract and/or order, except where otherwise indicated (see 6.2).

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4.3.1 Inspection conditions. Unless otherwise specified in this specification or applicable procurements documents (see 6.2), all inspections shall be performed in accordance with this specification and all the requirements of referenced documents.

4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Type I of MIL-STD-3064.

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table III. The methods of testing as specified wherever applicable and as listed in Table III shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yard full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirement specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE III. Material testing.

Characteristic	Requirement reference	Test method
Yarn ply	3.5.1	Visual
Weave	3.5.2	Visual
Labile sulfur	3.6.4	4.6.1
Colorfastness:		
Laundering (3 cycles)	Table I	AATCC TM61, Option 3A, 1/, 2/
Light	Table I	AATCC 16.3, Option 3, 3/
Perspiration	Table I	AATCC TM15, 2/
Crocking	Table I	AATCC TM8, 4/
Weight	Table II	ASTM D3776/D3776M
Yarns per inch	Table II	ASTM D3775
Breaking strength	Table II	ASTM D5034
Identification of mercerization	3.8	Microscopic examination for swelling of cotton fiber. 5/
pH	3.9	AATCC TM81
Dimensional stability	3.10	AATCC 96, Test Vc, D
Seam efficiency	3.11	ASTM D1683, 6/
Toxicity	3.12	4.6.3

1/ Only the stain on the nylon and cotton fibers of the color transfer cloth shall be evaluated.
2/ Rated using the AATCC EP1, Evaluation Procedure for Gray Scale for Color Change and AATCC EP2, Evaluation Procedure for Gray Scale for Staining.
3/ Rated using the AATCC EP1, Evaluation Procedure for Gray Scale for Color Change.
4/ Rated using the AATCC EP8, Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale.
5/ One determination shall be made from each sample unit and the result reported as “pass” or “fail”.
6/ The thread shall conform to Type I (soft finish) of A-A-52094. The needle size shall be 0.049 (± 0.001)-inch. The top (needle) thread shall be Tex 60, 3 ply, and the bottom (looper) thread shall be Tex 40, 3 ply.

4.6 Methods of testing and inspection.

4.6.1 Presence of labile sulfur test. Two (2) 1.50 (± 0.01) gram samples from each sample unit shall be tested. Each of the two (2) samples shall be cut into very small pieces and placed into separate test tubes. Prepare an acidic tin(II) chloride (stannous chloride) solution by dissolving 100 g $SnCl_2$ (ACS grade) in 100 mL of 35% hydrochloric acid (ACS grade) and adding the mixture to 50 mL of distilled water. Prepare a lead acetate solution by dissolving 5.0 g of lead acetate (CP reagent grade) in a small amount of distilled water and diluting to a total of 100 mL. If the solution is not clear, add a few drops (one at a time) of glacial acetic acid until the solution is clear. The samples shall be submersed the acidic tin(II) chloride solution. A filter paper wet out with the lead acetate solution shall be placed over the top of the test tube. The test tube containing the textile sample, tin(II) chloride and wet filter paper shall be heated over a low

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flame until the solution is boiling. The solution should not be heated for more than 15 seconds. A brown to black stain on the filter paper should be evaluated as follows:

Free - The filter paper shows no discoloration or staining of any kind.
Slight - The filter paper shows a light tan to light brown discoloration stain.
Moderate - The filter paper shows a dark brown discoloration stain.
Severe - The filter paper shows a black color stain.

The rating shall be recorded. The results shall be recorded as "pass" or "fail.

4.6.2 Visual shade matching. The color and appearance of the cloths shall match the standard sample when viewed using the AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles, Option C (see 6.4), with sources simulating artificial daylight D_{75} illuminant with a color temperature of 7500 (± 200) K illumination of 100 (± 20) foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2856 (± 200) K.

4.6.3 Toxicity test. When required, (see 6.2) an acute dermal irritation study and a skin sensitization study shall be conducted. When the results of these studies indicate the material is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (see 2.3). If toxicity requirement (see 3.12) can be demonstrated with historical use data, on the finishing treatments used, toxicity testing may not be required (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in clothing and equipage items for Military personnel.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Class required (see 1.2)
c. The specific issue of individual documents referenced (see 2.2)
d. Color required (3.6)
e. When first article is required (see 3.1)
f. Conformance inspection acceptance quality limits (AQLs) (see 4.3)
g. When toxicity testing is required (see 3.12, 4.6.3)
h. Width of cloth required (see 3.14)
i. Length required if other than specified (see 3.15)
j. Inspection conditions (see 4.3.1)
k. Packaging (see 5.1)

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Visual shade matching. In 2017, Option A of AATCC EP9, Evaluation Procedure for Visual Assessment of Color Difference of Textiles was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.6 Subject term (key word) listing.

Bag, barracks
Clothing
Equipage
Hood, chemical protective
Liner, shirt

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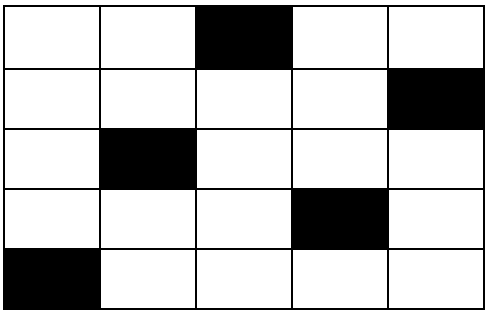


FIGURE 1. Weave.

Custodians:
Army - GL
Navy - NU
Air Force - 11

Preparing activity:
DLA-CT

(Project 8305-2021-011)

Review activities
Army - MD
Navy - CG1, MC

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